ANALYSTLAB AFRICA

Machine Learning Internship Programme · Week 2

Business Understanding Report

ShopSmart Retail Group — Order Value & Returns Analytics

Prepared by Umar Muhammad · ShopSmart Retail Group dataset

1. Executive summary

ShopSmart Retail Group operates 25 stores across Nigeria and sells both in-store and online. This report sets out the business problem behind the machine-learning work, how the data supports it, what the preprocessing revealed, and what the company can realistically expect from models built on this data.

Headline findings. The dataset is structurally complete (no missing values across 9 tables) but contains logical flaws that would mislead a model if left untreated. After preprocessing, 50,000 orders are ready for modelling. The key business conclusions are: (1) product returns (4.88% of orders) cannot be predicted from the information currently recorded, and (2) order value can only be "predicted" through arithmetic on quantity and price, so the high accuracy seen in Week 1 reflected a formula, not intelligence.

2. Business context and objective

Retail margins are sensitive to two things the business can influence: how much customers spend per order, and how often goods come back. Returns add reverse-logistics cost, refunds and lost stock; unreliable value forecasts make revenue planning harder.

Business problem	ML task	Target	Business value if solved
How much is an order worth?	Regression	Order_Value	Revenue forecasting, store targets, promotion planning
Will an order be returned?	Classification	Is_Returned	Flag risky orders early, reduce refund and logistics cost

Objective of Week 2: prepare one clean, documented, leakage-aware dataset so that Week 3 models are trained on trustworthy inputs, and report honestly on whether the data can answer the two questions above.

3. Stakeholders and how they would use the results

- **Store managers:** order-value expectations and return-risk flags for daily operations.
- **Finance team:** revenue forecasts; correct treatment of cancelled and pending orders.
- **Supply-chain and customer service:** early warning on likely returns, damaged or wrong items.
- **Data owner / IT:** fixing the recording problems found in this report (Section 5).

4. Data available

Nine tables describe the business. Six feed the order-level modelling table; three describe the organisation and stock rather than individual orders and were inspected but not joined.

Table	Rows	Role
Orders	50,000	Core table: one row per order (date, channel, status, price, quantity)
Order items	50,000	Line totals (source of Order_Value)
Customers	5,000	Age, gender, state, loyalty segment, registration date
Products	500	Category, cost price, catalogue selling price
Stores	25	Region, state, opening date
Returns	2,500	Return records (collapsed to one flag per order)
Employees / Inventory / Suppliers	125 / 500 / 489	Inspected only; not order-level

5. Data-quality issues with business impact

No cell was missing and no row was duplicated, yet the logic audit found problems that matter commercially:

- **Discount is applied as a flat number.** Order totals are Quantity × Price minus a 0–20 figure that looks like a percentage. If the discount is meant to be a percentage, recorded order values are overstated by about 15,120 per order on average. This should be confirmed with the data owner.
- **Returns do not follow the business flow.** 67.4% of returns are attached to orders that were cancelled or still pending, and 50.1% carry a return date earlier than the purchase date.
- **A third of orders are cancelled and a third pending.** Only about 33% of orders are Completed, so any revenue figure must be filtered by status.
- **Reference data is inconsistent.** Order prices never match catalogue prices; six of ten states appear under several store regions; store names repeat across cities.
- **Customer data quality.** 10.3% of orders pre-date the customer's registration, and 49 customer records repeat an e-mail address already used by another customer.

6. What the preprocessing concluded

The questions below were set in the assignment; each answer is supported by the evidence in the Data Preprocessing Report.

Business question	Answer
What data-quality issues exist?	13 logic issues despite complete data (Section 5).
Which features need preprocessing?	Dates stored as text, 8 categorical columns, skewed money columns, personal data, and the returns table.
Which encoding fits?	Ordinal for segment and age band; label for 2-level columns; one-hot for 5 nominal columns.
Which scaling?	Robust for skewed, MinMax for bounded, Standard for symmetric columns; fitted on training rows only.
Significant outliers?	410 high-value baskets (0.82%) flagged but genuine; kept, with robust

Business question	Answer
	scaling to limit influence.
Which features matter most?	For returns: none meaningfully (ROC-AUC 0.4955, i.e. chance). For order value: quantity and price, which are the formula.
Ready for model training?	Technically yes (50,000 × 40, clean and numeric); substantively limited, see Section 7.

7. Business implications and recommendations

- **Do not promise a returns-prediction model yet.** With the current fields, a model scores ROC-AUC 0.4955 (random guessing is 0.50). Return rates sit between about 4.1% and 5.7% in every segment examined. Useful prediction would need new data: product-level return history, delivery delays, product defects, customer reviews.
- **Treat high order-value accuracy with suspicion.** A model given quantity and price reaches R^2 = 1.00 because it recomputes the formula; without those inputs R^2 falls to essentially zero (-0.002). Agree what is known at prediction time before judging model skill.
- **Fix recording at source.** Confirm whether Discount is a percentage, enforce return dates after order dates, and block returns on cancelled orders.
- **Report with the right metrics.** Because only 4.88% of orders are returned, accuracy is misleading; use ROC-AUC, precision, recall and the precision-recall curve.
- **Protect personal data.** Names, e-mails and phone numbers were removed from the modelling files.

8. Success criteria for Week 3

- Models evaluated on the stored test split, never on training rows.
- Classification judged against a majority-class baseline using recall and ROC-AUC, not accuracy.
- Regression reported separately with and without the arithmetic inputs, and the difference explained.